D 11802 **(Pages : 3)** **Name.....**

Reg. No.....

THIRD SEMESTER (CUCBCSS–UG) DEGREE EXAMINATION
NOVEMBER 2021

B.Sc. L.R.P. (Alternate Pattern)

A11—BASIC NUMERICAL SKILLS

(2017—2018 Admissions)

Time : Three Hours Maximum : 80 Marks

Part A

Answer **all** *questions.*
Each question carries 1 *mark.*

I. Choose the correct answer :

1 A set 'A' is said to be a subset of a set B if every element of A is :

(a) Also an element of A. (b) Also an element of B.

(c) Not an element of B. (d) Not an element of A .

2 A ———————— is matrix which has only one column.

(a) Column. (b) Row.

(c) Diagonal. (d) Square.

3 The equations $x^2 + 4 = 0$ is a ————————.

(a) Quadratic equation. (b) Cubic equation.

(c) Simple equation. (d) None.

4 A series obtained by adding a constant number to its preceding terms is :

(a) GP. (b) AP.

(c) GP or AP. (d) None.

5 The primary data are collected by :

(a) Interview. (b) Observation.

(c) Questionnaire. (d) All these.

Turn over

II. Fill in the blanks :

6 Find the 7th term of series 1, 4, 7 ————————.

7 A (B + C) = AB + AC is a ———————— law.

8 $X^2 - 4 = 0$ implies x =.....

9 ———————— is filled by the enumerator to collect primary data.

10 When mean is less than median and median is less than mode the distribution is said to be ————————.

(10 × 1 = 10 marks)

Part B

Answer any **eight** *questions.*
Each question carries 2 *marks.*

11 Define EQUAL SET and EQUIVALENT SET.

12 If A ={1, 3, 5, 7 }, B = { 5, 9, 13, 17 }, C = {1, 3, 9, 13 }, find :

(a) $A \cap B$; and $(A - B) - C$.

13 Solve $4(x - 2) + 5(x - 3) - 25 = x + 8$.

14 Two third of a number decreased by 2 equals 4. Find the number.

15 What is a geometric progression ?

16 Find the 12th term of an A.P. 6, 2, – 2.

17 What is tabulation ? What are the objectives ?

18 Calculate median : 25, 35, 15, 18, 17, 36, 28, 24, 22, 26.

19 What are the relative measures of dispersion or variability of data ?

20 Find the simple interest and amount for Rs. 50,000 at 7.5% p. a for 4 months.

(8 × 2 = 16 marks)

Part C

Answer any **six** *questions.*
Each question carries 4 *marks.*

21 In a college there are 20 teachers, who teach Accountancy or Statistics. Of these 12, teach Accountancy and 4 teach both Statistics and Accountancy. How many teach Statistics ?

22 Find the value of a, b if :

$$2 \times \begin{bmatrix} a & 5 \\ 7 & b-3 \end{bmatrix} + \begin{bmatrix} 3 & -4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ 15 & 14 \end{bmatrix}.$$

23 Solve $3x + 4y = 7$, and $4x - 7 = 3$.

24 Find many two digit numbers are divisible by 3 ?

25 At what rate would a sum of money double in 20 years ?

26 Draw a Pie diagam for the following data :

Prime Cost	... 30%
Factory over Head	... 18%
Administrative overhead	... 28%
Selling and Distribution overhead	... 14%
Profit	... 10%

27 What is skewness ? What are the types of skewness ?

28 Find the H.M. of the series : 2, 3, 4, 5.

(6 × 4 = 24 marks)

Part D

Answer any **two** *questions.*
Each question carries 15 *marks.*

29 Find four numbers of A.P. whose sum is 20 and the sum of whose square is 120.

30 Following are the data related with the prices and quantities consumed for 2010 and 2012. Construct price index numbers by : (1) Laspeyre's method ; (2) Paasche's method ; (3) Bowly's - Dorbish method ; and (4) Fisher's method.

Commodity	2010		2012	
	Price	*Quantity*	*Price*	*Quantity*
Rice	5	15	7	12
Wheat	4	5	6	4
Sugar	7	4	9	3
Tea	52	2	55	2

31 Compare primary data and secondary data. What is a questionnaire ? What are the essentials of a good questionnaire.

(2 × 15 = 30 marks)